

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 03/04/2020

Quiz 6: Section 3.5, 3.6, 3.7

Name (Print): _____

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Section 3.5

2 pts

1. Find $\frac{dy}{dx}$ given that $y = \sec(x) \csc(x)$.

- 2 pts 2. Find $\frac{dy}{dx}$ given that $y = \frac{\sin(x) + 1}{\cot(x)}$.

Section 3.6

3 pts

1. Suppose a stone is thrown vertically upward with an initial velocity of 64 ft/s from a bridge 96 ft above a river. By Newton's laws of motion, the position of the stone (measured as the height above the river) after t seconds is

$$s(t) = -16t^2 + 64t + 96,$$

where $s = 0$ is the level of the river. Approximately with what velocity will the stone strike the river?

Section 3.7

- 3 pts 1. Find $\frac{d}{dx} \left[(\tan(x) + 10)^{21} \right]$.